ITA0605 - Machine Learning
List of Lab Exercises

2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Aim:

To implement and demonstrate the Candidate-Elimination Algorithm using a CSV file to find the set of all hypotheses consistent with the given training examples.

Algorithm:

1. Import the required libraries (`pandas`).
2. Read the training data from a CSV file.
3. Initialize the Specific hypothesis (S) with the first positive example.
4. Initialize the General hypothesis (G) with the most general hypothesis (`?`).
5. For each training example:
    - If it is positive, generalize the Specific hypothesis.
    - If it is negative, specialize the General hypothesis.
6. Remove inconsistent hypotheses.
7. Display the final Specific (S) and General (G) hypotheses.

Dataset 1:

Program:

```
import pandas as pd

import numpy as np

sample_data = [

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Warm', 'High',   'Strong', 'Warm', 'Same', 'Yes'],

    ['Rainy', 'Cold', 'High',   'Strong', 'Warm', 'Change', 'No'],

    ['Sunny', 'Warm', 'High',   'Strong', 'Cool', 'Same', 'Yes'],

    ['Rainy', 'Warm', 'Normal', 'Weak',   'Warm', 'Same', 'No'],

    ['Sunny', 'Warm', 'Normal', 'Weak',   'Warm', 'Same', 'Yes'],

    ['Cloudy', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Cold', 'High',   'Weak',   'Cool', 'Change', 'No']
```

```
]

column_names = [
    'Sky',
    'Temperature',
    'Humidity',
    'Wind',
    'Water',
    'Forecast',
    'PlayTennis'
]
data = pd.DataFrame(sample_data, columns=column_names)

print("Play Tennis Dataset:")
print(data)

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]
for i, h in enumerate(concepts):

    if target[i] == "Yes":
        for x in range(len(S)):
```

```
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"

    else:
        for x in range(len(S)):

            if h[x] != S[x]:
                G[x][x] = S[x]

            else:
                G[x][x] = "?"
G = [
    g for g in G
    if g != ["?" for i in range(len(S))]
]

print("\n-----------------------------------")
print("Candidate Elimination Results")
print("-----------------------------------")

print("\nSpecific Hypothesis (S):")
print(S)

print("\nGeneral Hypothesis (G):")
```

for g in G:

print(g)

Output:

```
import pandas as pd
import numpy as np
sample_data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
column_names = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport']
data = pd.DataFrame(sample_data, columns=column_names)
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]
for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"
    else:
        for x in range(len(S)):
            if h[x] != S[x]:
                G[x][x] = S[x]
            else:
                G[x][x] = "?"
G = [g for g in G if g != ["?" for i in range(len(S))]]
print("Specific Hypothesis:")
print(S)
print("\nGeneral Hypothesis:")
for g in G:
    print(g)
```

```
Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Dataset 2:

Program:

```
import pandas as pd
import numpy as np

sample_data = [
['High','Good','Permanent','Yes','Young','Yes'],
['High','Good','Permanent','No','Middle','Yes'],
['Low','Poor','Temporary','No','Young','No'],
['Medium','Good','Permanent','Yes','Middle','Yes'],
['High','Average','Temporary','Yes','Old','No'],
['Medium','Good','Permanent','No','Young','Yes'],
['Low','Good','Permanent','Yes','Middle','Yes'],
['Low','Poor','Temporary','Yes','Old','No']
]
```

```
column_names = ['Income','CreditScore','Employment','Property','Age','LoanApproved']

data = pd.DataFrame(sample_data, columns=column_names)

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]

for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"
    else:
        for x in range(len(S)):
            if h[x] != S[x]:
                G[x][x] = S[x]
            else:
                G[x][x] = "?"

G = [g for g in G if g != ["?" for i in range(len(S))]]

print("Specific Hypothesis:")
print(S)

print("\nGeneral Hypothesis:")
for g in G:
print(g)
```

Output:

```
import pandas as pd
import numpy as np

sample_data = [
['High','Good','Permanent','Yes','Young','Yes'],
['High','Good','Permanent','No','Middle','Yes'],
['Low','Poor','Temporary','No','Young','No'],
['Medium','Good','Permanent','Yes','Middle','Yes'],
['High','Average','Temporary','Yes','Old','No'],
['Medium','Good','Permanent','No','Young','Yes'],
['Low','Good','Permanent','Yes','Middle','Yes'],
['Low','Poor','Temporary','Yes','Old','No']
]

column_names = ['Income','CreditScore','Employment','Property','Age','LoanApproved']

data = pd.DataFrame(sample_data, columns=column_names)

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]

for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"
```

```
        else:
            for x in range(len(S)):
                if h[x] != S[x]:
                    G[x][x] = S[x]
                else:
                    G[x][x] = "?"

    G = [g for g in G if g != ["?" for i in range(len(S))]]

    print("Specific Hypothesis:")
    print(S)

    print("\nGeneral Hypothesis:")
    for g in G:
        print(g)
```

```
Specific Hypothesis:
['?' 'Good' 'Permanent' '?' '?']

General Hypothesis:
['?', 'Good', '?', '?', '?']
['?', '?', 'Permanent', '?', '?']
```

Dataset 3:

Program:

```
import pandas as pd
import numpy as np

sample_data = [
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']
]

column_names = [
    'CGPA',
    'Communication',
    'Internship',
    'Programming',
    'Aptitude',
    'Placed'
```

```
]
data = pd.DataFrame(sample_data, columns=column_names)

print("Student Placement Dataset:")
print(data)

concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        # Positive example
        for x in range(len(S)):

            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"

    else:

        # Negative example
        for x in range(len(S)):

            if h[x] != S[x]:
                G[x][x] = S[x]

            else:
                G[x][x] = "?"

G = [
    g for g in G
    if g != ["?" for i in range(len(S))]
]

print("\n-----------------------------------")
print("Candidate Elimination Results")
print("-----------------------------------")
```

```
print("\nSpecific Hypothesis (S):")
print(S)

print("\nGeneral Hypothesis (G):")
for g in G:
print(g)
```

Output:

```
Student Placement Dataset:
     CGPA Communication Internship Programming Aptitude Placed
0    High          Good        Yes        Good     High    Yes
1    High     Excellent        Yes        Good     High    Yes
2  Medium       Average         No     Average   Medium     No
3    High          Good        Yes   Excellent     High    Yes
4     Low          Poor         No     Average      Low     No
5    High          Good        Yes        Good   Medium    Yes

-----------------------------------
Candidate Elimination Results
-----------------------------------

Specific Hypothesis (S):
['High' '?' 'Yes' '?' '?']

General Hypothesis (G):
['High', '?', '?', '?', '?']
['?', '?', 'Yes', '?', '?']
```

Dataset 4:

Program:

```
import pandas as pd
import numpy as np

sample_data = [
   ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
   ['Yes', 'Yes', 'No',  'Yes', 'Yes', 'Positive'],
   ['No',  'Yes', 'Yes', 'No',  'No',  'Negative'],
   ['Yes', 'Yes', 'Yes', 'No',  'Yes', 'Positive'],
   ['No',  'No',  'Yes', 'Yes', 'No',  'Negative'],
   ['Yes', 'Yes', 'No',  'No',  'Yes', 'Positive'],
   ['Yes', 'No',  'Yes', 'Yes', 'Yes', 'Positive'],
   ['No',  'No',  'No',  'No',  'No',  'Negative']
]
```

```
column_names = [
    'Fever',
    'Cough',
    'Headache',
    'BodyPain',
    'Fatigue',
    'Disease'
]
data = pd.DataFrame(sample_data, columns=column_names)

print("Disease Diagnosis Dataset:")
print(data)



concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]

for i, h in enumerate(concepts):

    if target[i] == "Positive":
        for x in range(len(S)):

            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"

    else:
        for x in range(len(S)):

            if h[x] != S[x]:
                G[x][x] = S[x]

            else:
                G[x][x] = "?"

G = [
    g for g in G
    if g != ["?" for i in range(len(S))]
```

```
]


print("\n----------------------------------")
print("Candidate Elimination Results")
print("----------------------------------")

print("\nSpecific Hypothesis (S):")
print(S)

print("\nGeneral Hypothesis (G):")
for g in G:
print(g)
```

Output:

```
Disease Diagnosis Dataset:
  Fever Cough Headache BodyPain Fatigue   Disease
0   Yes   Yes      Yes      Yes     Yes  Positive
1   Yes   Yes       No      Yes     Yes  Positive
2    No   Yes      Yes       No      No  Negative
3   Yes   Yes      Yes       No     Yes  Positive
4    No    No      Yes      Yes      No  Negative
5   Yes   Yes       No       No     Yes  Positive
6   Yes    No      Yes      Yes     Yes  Positive
7    No    No       No       No      No  Negative

----------------------------------
Candidate Elimination Results
----------------------------------

Specific Hypothesis (S):
['Yes' '?' '?' '?' 'Yes']

General Hypothesis (G):
['Yes', '?', '?', '?', '?']
['?', '?', '?', '?', 'Yes']
```

Dataset 5:

Program:

```
import pandas as pd
import numpy as np

sample_data = [
    ['Yes', 'Yes', 'Yes', 'No',  'Yes', 'Yes'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],
    ['No',  'No',  'No',  'No',  'No',  'No'],
    ['Yes', 'No',  'Yes', 'No',  'Yes', 'Yes'],
    ['No',  'Yes', 'No',  'Yes', 'No',  'No'],
    ['Yes', 'Yes', 'Yes', 'No',  'No',  'Yes'],
    ['Yes', 'No',  'Yes', 'Yes', 'Yes', 'Yes'],
    ['No',  'No',  'Yes', 'No',  'No',  'No']
]

column_names = [
    'ContainsLink',
    'ContainsOffer',
    'UnknownSender',
    'Attachment',
    'ManyRecipients',
    'Spam'
]
data = pd.DataFrame(sample_data, columns=column_names)

print("Email Spam Detection Dataset:")
print(data)



concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
S = concepts[0].copy()
G = [["?" for i in range(len(S))] for j in range(len(S))]
for i, h in enumerate(concepts):

    if target[i] == "Yes":
        for x in range(len(S)):

            if h[x] != S[x]:
```

```
            S[x] = "?"
            G[x][x] = "?"

    else:
        for x in range(len(S)):

            if h[x] != S[x]:
                G[x][x] = S[x]

            else:
                G[x][x] = "?"

G = [
    g for g in G
    if g != ["?" for i in range(len(S))]
]

print("\n-----------------------------------")
print("Candidate Elimination Results")
print("-----------------------------------")

print("\nSpecific Hypothesis (S):")
print(S)

print("\nGeneral Hypothesis (G):")
for g in G:
print(g)
```

Output:

```
...  Email Spam Detection Dataset:
       ContainsLink ContainsOffer UnknownSender Attachment ManyRecipients Spam
     0          Yes           Yes           Yes         No             Yes  Yes
     1          Yes           Yes           Yes        Yes             Yes  Yes
     2           No            No            No         No              No   No
     3          Yes            No           Yes         No             Yes  Yes
     4           No           Yes            No        Yes              No   No
     5          Yes           Yes           Yes         No              No  Yes
     6          Yes            No           Yes        Yes             Yes  Yes
     7           No            No           Yes         No              No   No

     -----------------------------------
     Candidate Elimination Results
     -----------------------------------

     Specific Hypothesis (S):
     ['Yes' '?' 'Yes' '?' '?']

     General Hypothesis (G):
     ['Yes', '?', '?', '?', '?']
```

Result:

The Candidate-Elimination algorithm was successfully implemented using a CSV file. It generated the Specific Hypothesis (S) and the General Hypothesis (G), representing the version space of all hypotheses that are consistent with the given training examples.